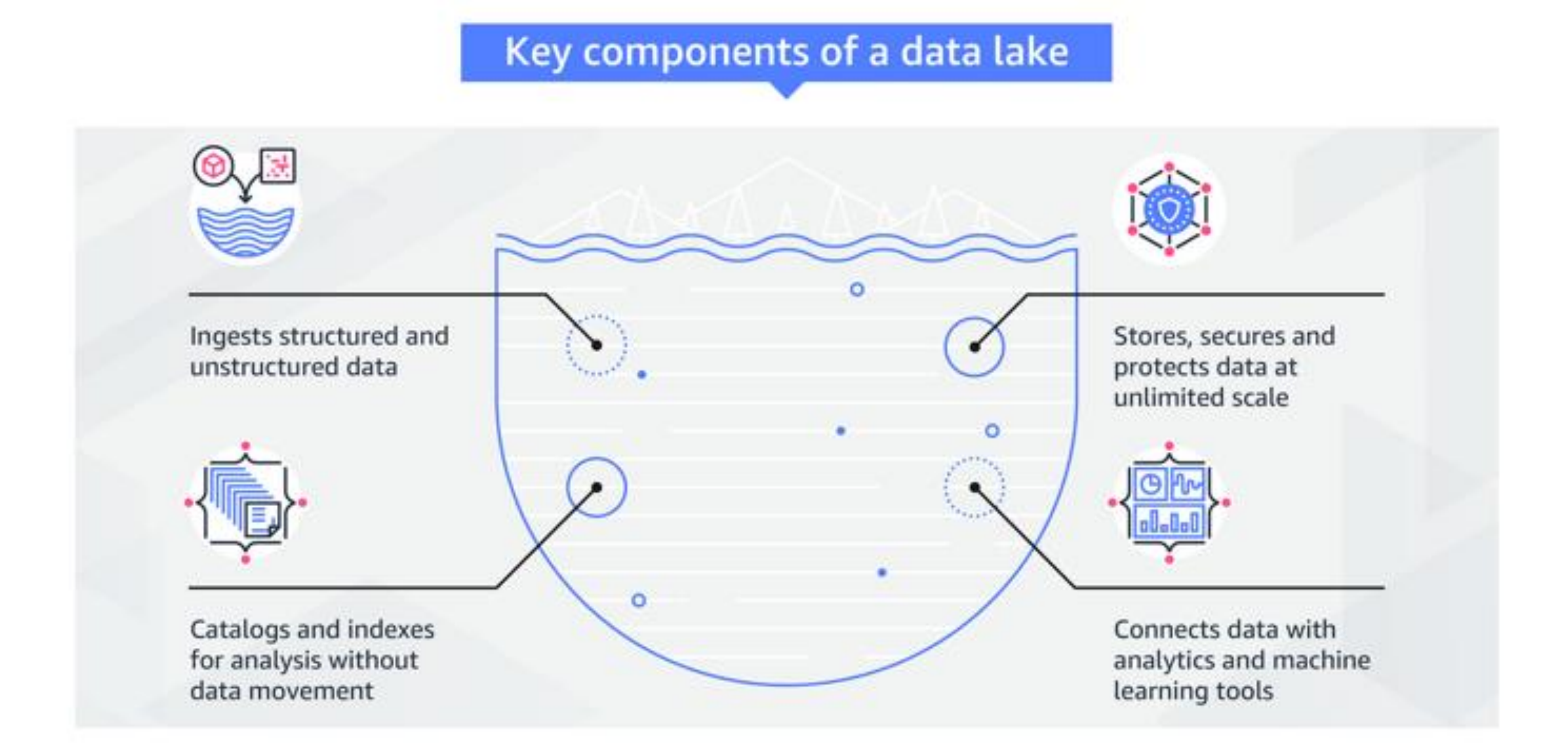
RECAP

- What is a primary advantage of a Data Lake compared to a traditional database?
  - A. Requires predefined structure before storage
  - B. Stores only structured data
  - C. Stores data as-is, without needing to define structure in advance
  - D. Cannot handle unstructured data
- Which of the following best describes the difference between a Data Lake and a Data Warehouse?
  - A. A Data Warehouse stores raw, unstructured data, while a Data Lake stores processed data
  - B. A Data Lake is limited to structured data, while a Data Warehouse handles unstructured data
  - C. A Data Warehouse stores processed, structured data, while a Data Lake stores raw, undefined data
  - D. Both are interchangeable storage systems for the same purpose
- Data Lakes require predefined schemas before data can be ingested
  - True
  - False

RECAP

- Big Data Analytics is the complex process of examining large and varied data sets, or big data, to uncover information such as hidden patterns, unknown correlations, market trends and customer preferences that can help organizations make informed business decisions
- Why important?
  - Cost Reduction
  - New products and services
  - Faster and smarter better decision
  - Time to market reductions

- A <u>Data Lake</u> allows you to store all your structured and unstructured data, in one centralized repository, and at any scale
- With a Data Lake, you store your data <u>as-is</u>, without having to first structure the data, based on potential questions you may have in the future



RECAP

- Spark SQL is a Spark module for structured data processing
- It provides a programming abstraction called DataFrames and can also act as a distributed SQL query engine
- It enables unmodified Hadoop Hive queries to run up to 100x faster on existing deployments and data
- It also provides powerful integration with the rest of the Spark ecosystem

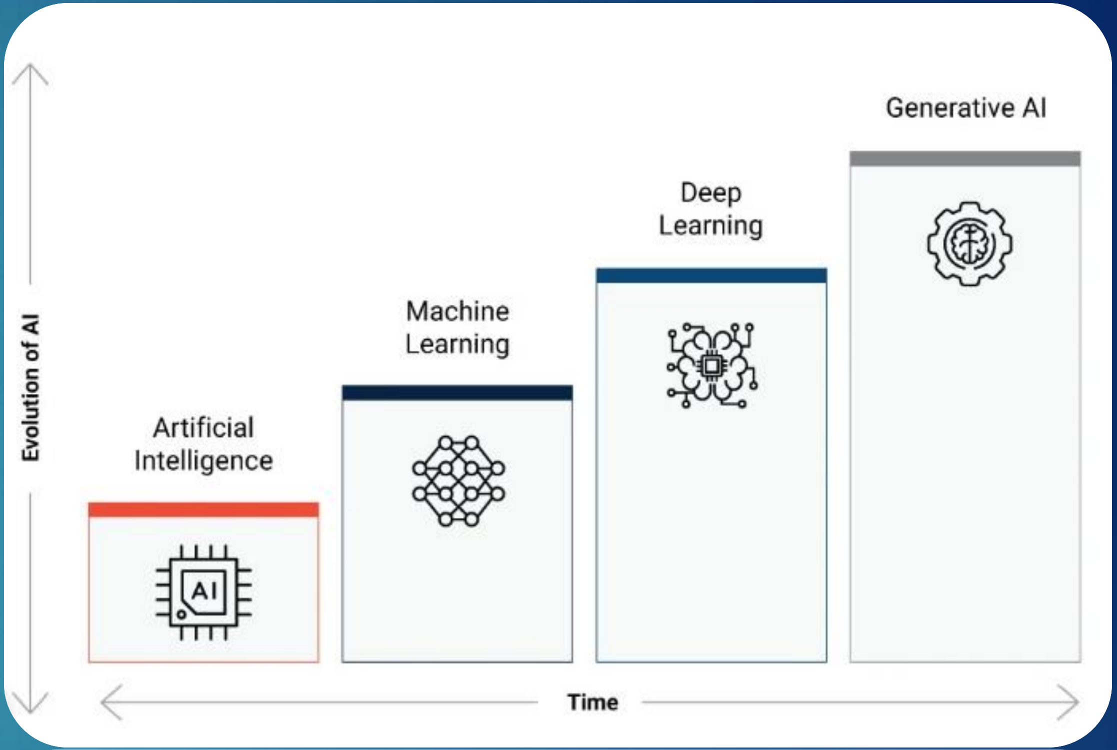
Generative AI on the Cloud

Engineering Cloud Software Systems

Department of Software Engineering
Rochester Institute of Technology

Overview

- Artificial Intelligence (AI) refers to the ability of machines to perceive, reason, learn, and make decisions
- Within AI, Machine Learning (ML), Deep Learning (DL), and Generative AI (GenAI) have emerged as major subfields
- Today, cloud platforms (AWS, Azure, Google Cloud) provide the scalable compute, storage, and APIs that make training and deploying these models feasible, lowering the barrier for adoption

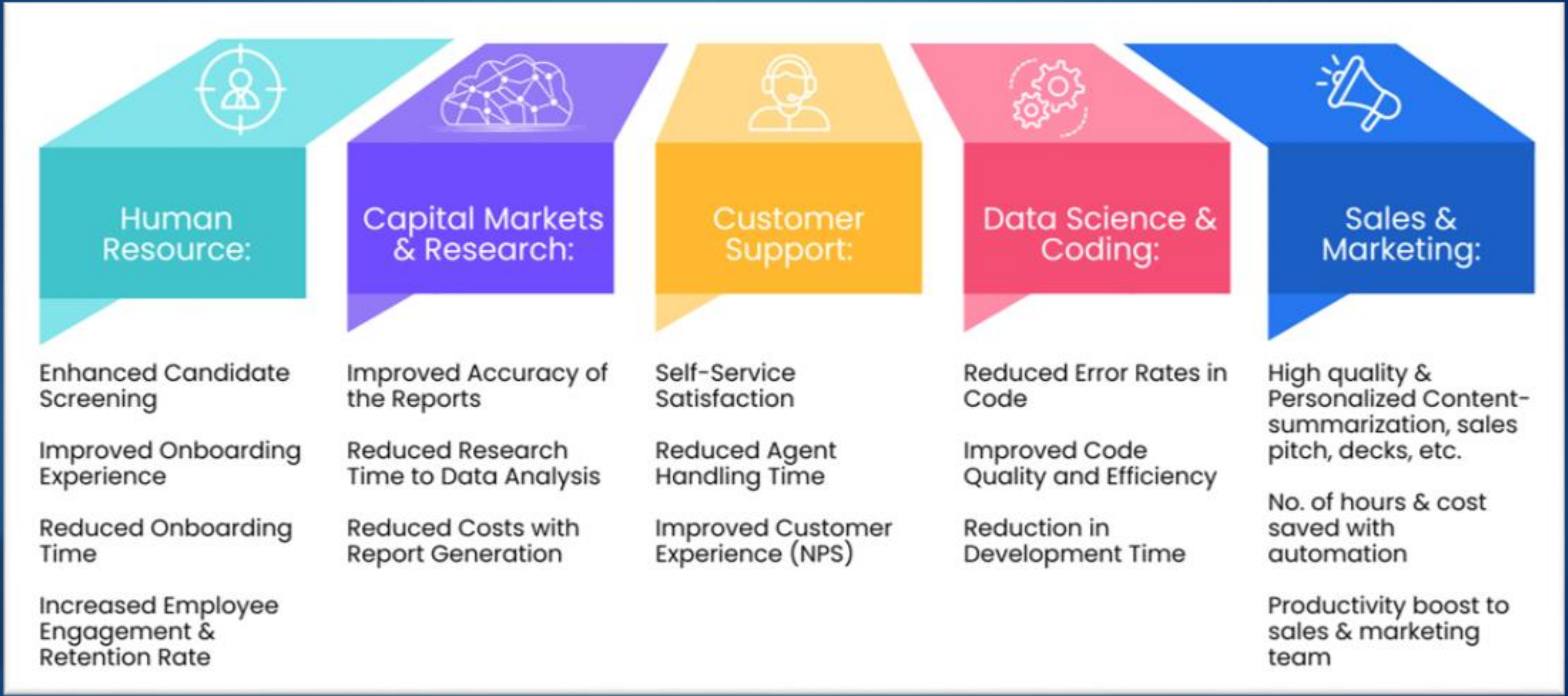


Source: https://synoptek.com/insights/it-blogs/data-insights/ai-ml-dl-and-generative-ai-face-off-a-comparative-analysis/

Traditional AI vs. Generative AI (GenAI)

	GenAI	Traditional AI
Purpose and Output	**Specifically designed to create new content**, such as text, images, music, or code, that resembles human-made creations	Used to analyze data, **recognize patterns, make predictions**, or automate tasks without necessarily generating new, creative outputs.
Learning Approach	Models, such as **foundation models**, are trained on vast datasets to learn the underlying patterns and structures of the data, enabling them to generate new content based on this learned information	Models may rely on more specialized, task-specific training and often focus on optimizing accuracy for specific tasks like classification or regression
Data Utilization	Leverages large-scale, diverse datasets to develop a **broad understanding** that can be applied to various creative tasks	Models are often trained on more narrowly defined datasets **tailored for specific problem-solving** or decision-making applications.
Interactivity and Adaptability	Models are designed to interact with users in **more creative ways**, producing outputs that can evolve based on user input and context	**More rigid**, focusing on predefined tasks and not necessarily designed for creative adaptability

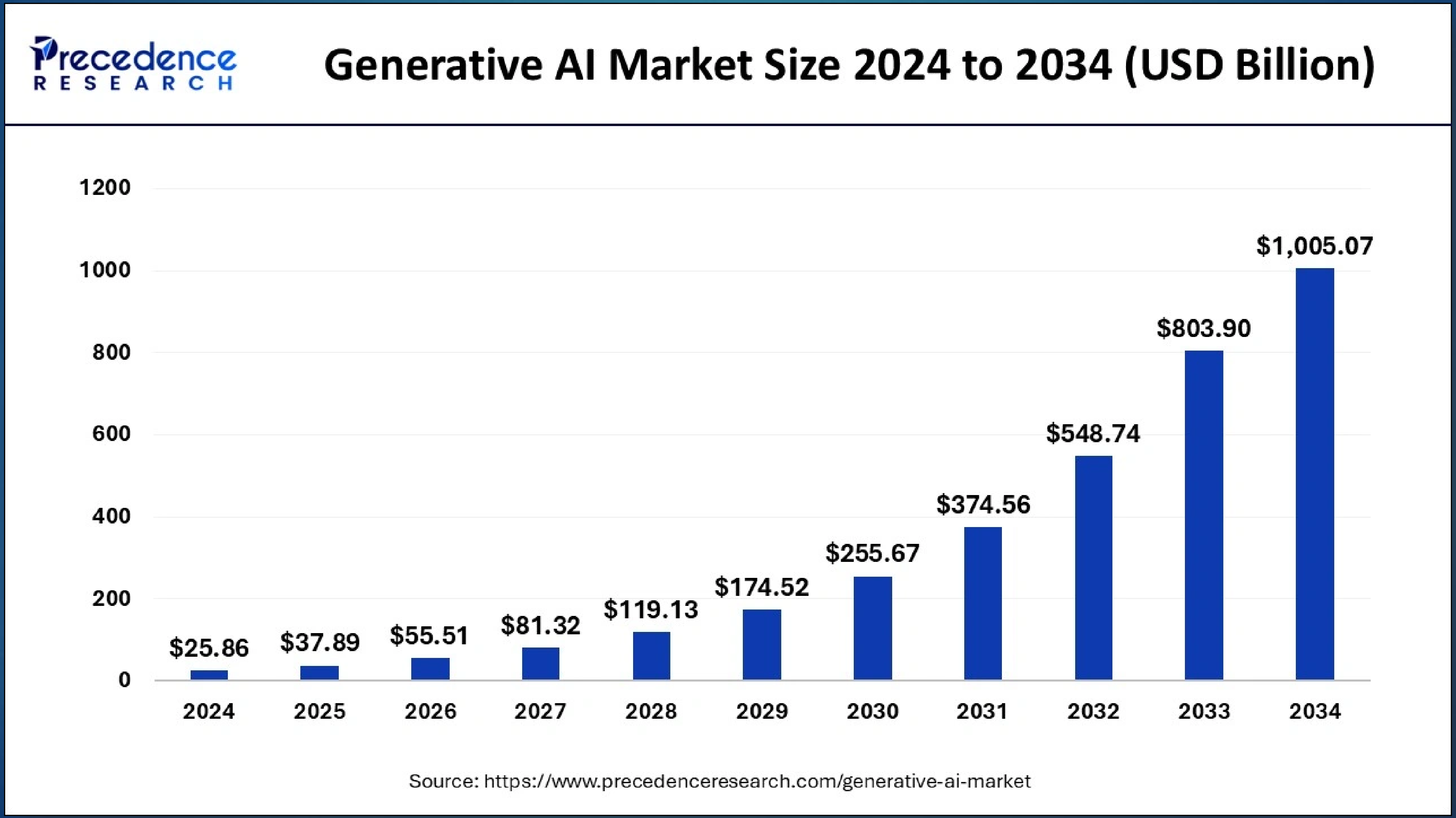
Common GenAI Use Cases



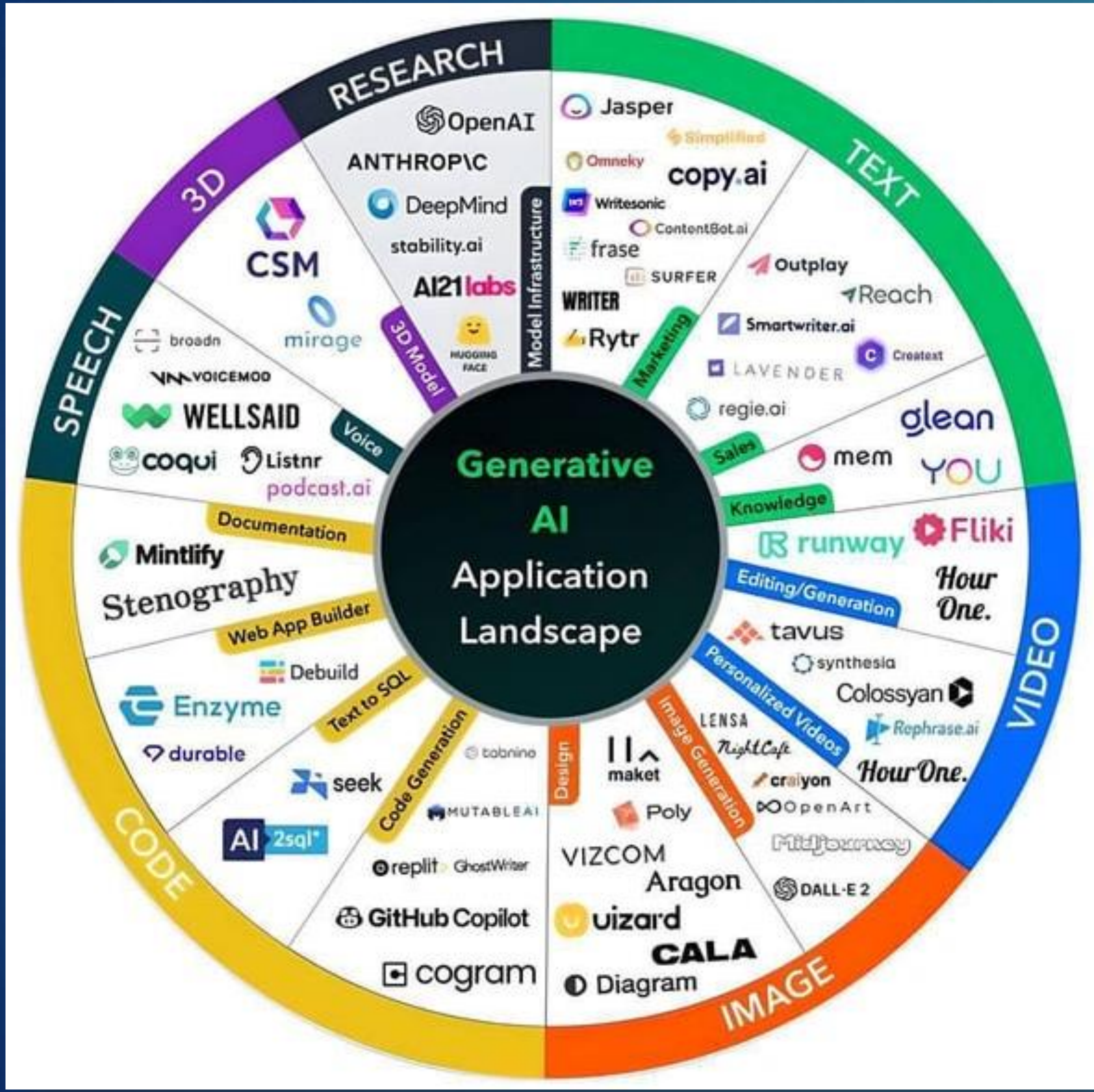
Source: https://www.fluid.ai/blog/how-do-you-measure-gen-ai-deployment-pilot-success-key-performance-indicators-and-metrics

GenAI Market Size and Growth

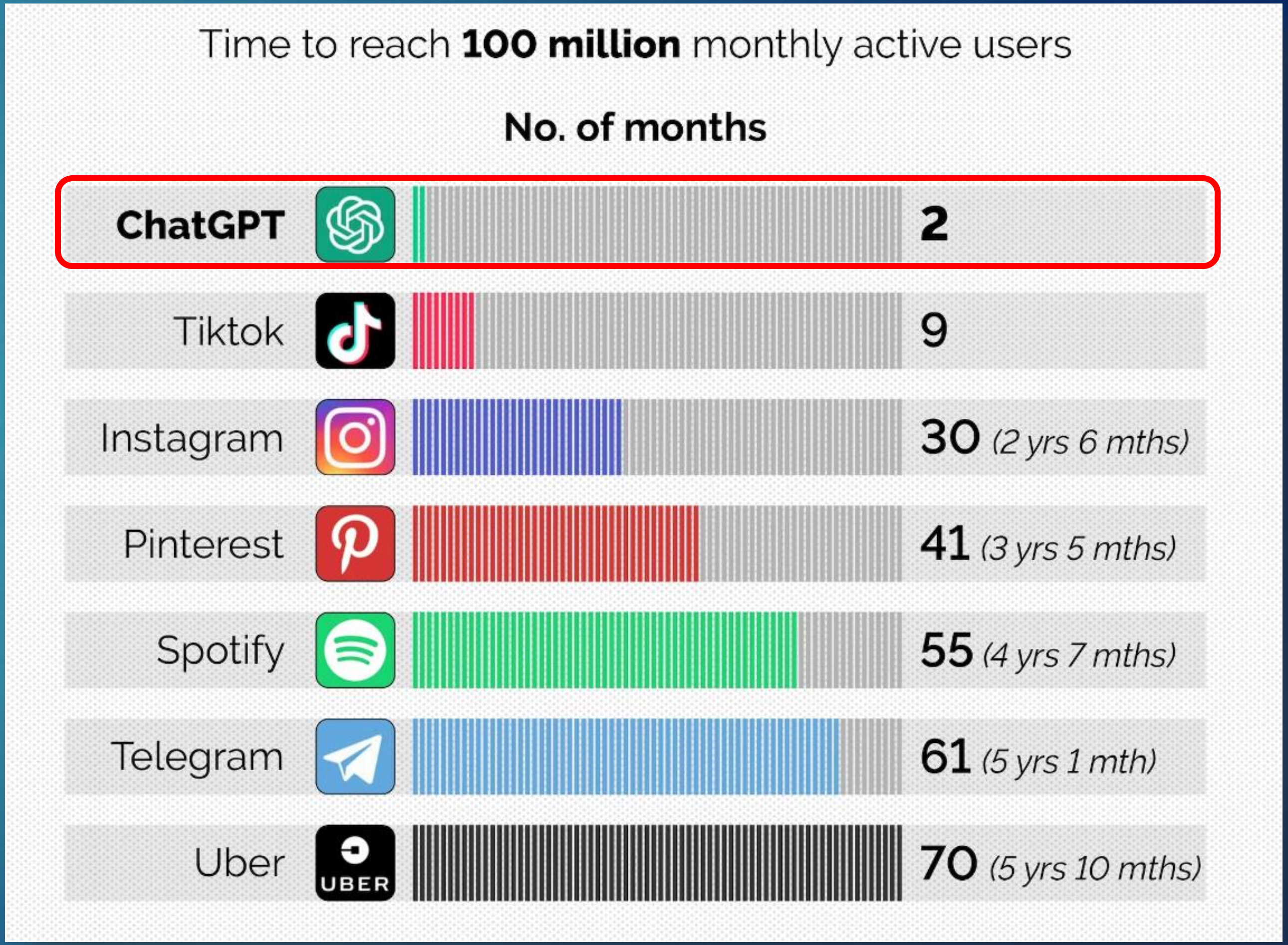
- Analyst estimates vary, but consensus shows rapid GenAI growth



GenAI Tools Popularity



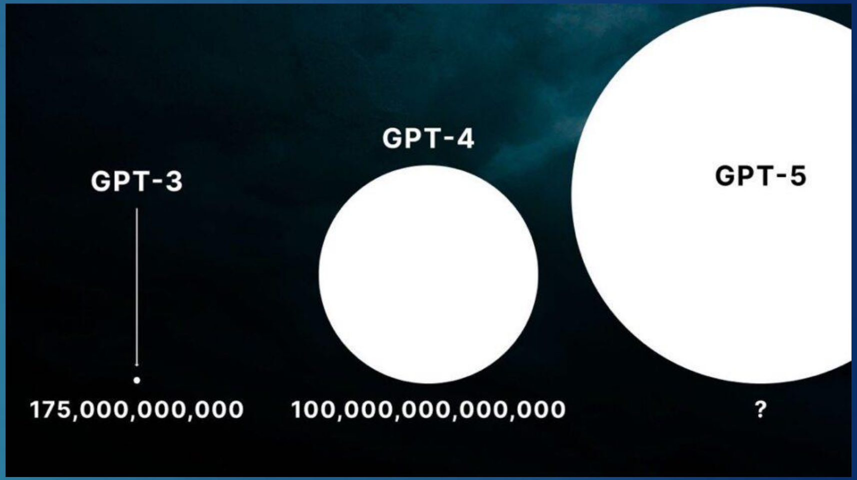
Source: https://www.reddit.com/r/ChatGPT/comments/13up0c6/ai_tools_apps_in_one_place_sorted_by_category/

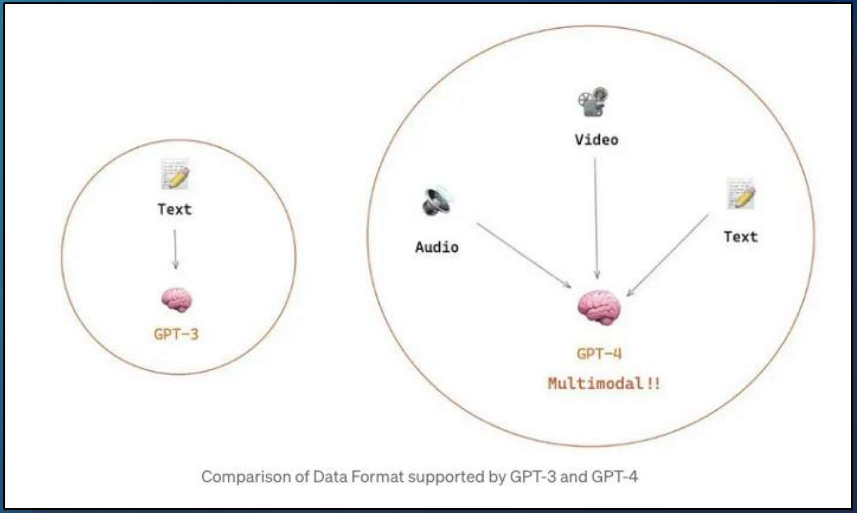


Source: https://medium.com/@rajabhinav/generative-ai-a-beginners-guide-e707c78cdc6f

ChatGPT Facts & Figures

- ChatGPT sits on top of Microsoft's Azure Cloud infrastructure, which charges around $3 per hour for a single A100GPU supercomputer cluster
- OpenAI reportedly used 1,023 A100 GPUs to train ChatGPT, so it is possible that the training process was completed in as little as 34 days
- Software industry experts have estimated that running costs for ChatGPT range somewhere between $100,000 and $700,000 per day or $3-$21 million per month
- The dataset for training ChatGPT-4 is estimated to be in the trillions of parameters and ChatGPT-5 in the quadrillions
- ChatGPT was primarily written in the Python computer programming language but is also capable of understanding and producing responses in a wide range of computer languages
- Operating costs are significant where estimates range from $100M–$1B annually depending on scale and usage





Comparison of Data Format supported by GPT-3 and GPT-4

Source: https://www.stylefactoryproductions.com/blog/chatgpt-statistics

Source: https://medium.com/@sanchit0496/whats-new-in-chatgpt4-how-is-it-better-than-chatgpt3-32b788a67f9e

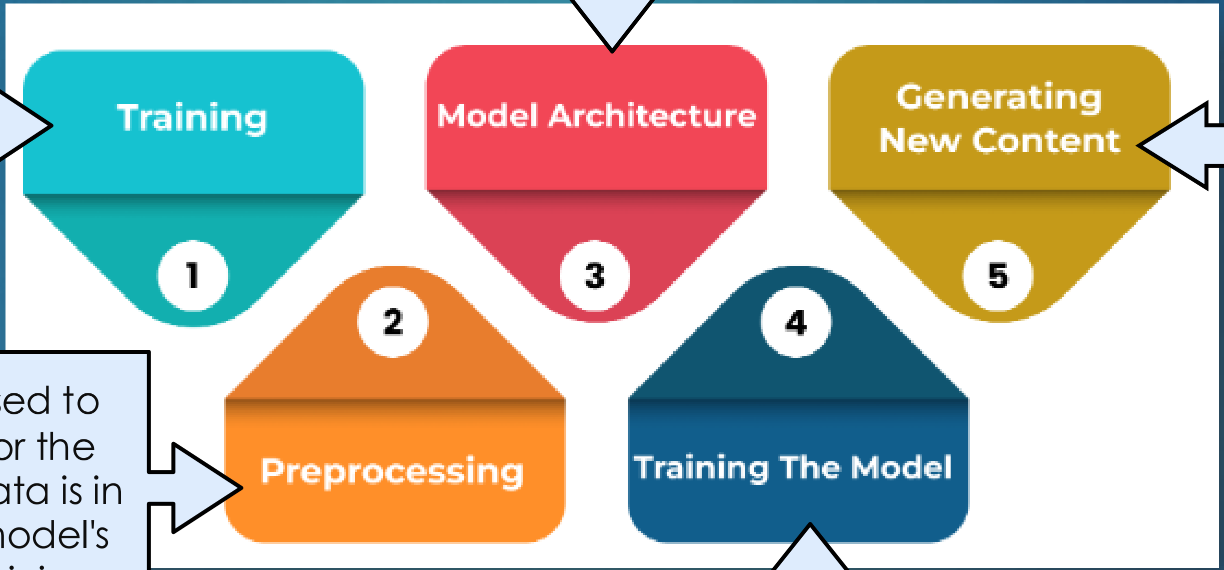
What has Caused GenAI to Gain Popularity?

- Advances in Model Architecture and Scale
  - The trend towards larger models trained on vast datasets has enabled generative AI to produce more accurate and creative outputs. OpenAI's GPT-3, for example, has 175 billion parameters, allowing it to generate high-quality text across various topics.
- Improved Computational Power
  - The increase in computational power, particularly with the advent of advanced GPUs (Graphics Processing Units) and TPUs (Tensor Processing Units), has made it feasible to train large-scale AI models.
  - Cloud-based platforms like AWS, Google Cloud, and Azure have made high-performance computing more accessible, allowing more organizations and researchers to develop and deploy GenAI models without investing in expensive hardware
- Open-Source Models and Tools
  - The release of powerful pre-trained models, such as GPT, DALL-E, Stable Diffusion, and BERT, as open-source or through API access, has increased access to advanced AI capabilities. Developers and researchers can now use these models as a base for various applications without the need for extensive training resources.

How Does GenAI work?

The model architecture defines the structure and complexity of the neural network used in GenAI. It includes layers and connections that enable the model to capture intricate relationships and generate high-quality content.

GenAI begins by training models on large and diverse datasets, allowing the model to learn a broad range of patterns and features. This training involves using algorithms to optimize the model's ability to predict and generate data.



Once trained, the model uses its learned patterns to generate new content by sampling from its learned distributions. This output is crafted to be coherent and relevant, mimicking the patterns seen during training.

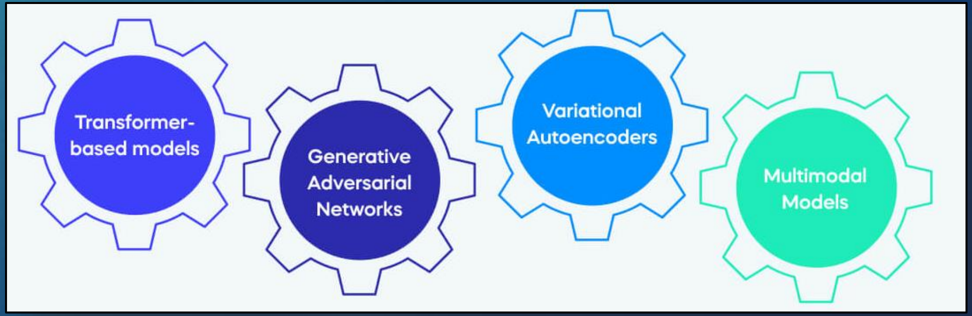
Before training, data is pre-processed to clean, normalize, and structure it for the model. This step ensures the input data is in a suitable format, enhancing the model's efficiency and accuracy during training.

During this phase, the model is trained using the pre-processed data, adjusting its parameters to minimize errors and improve its generative capabilities. This involves iterative adjustments based on feedback to refine the model's performance.

Source: https://www.solulab.com/generative-ai/

Types of GenAI Models

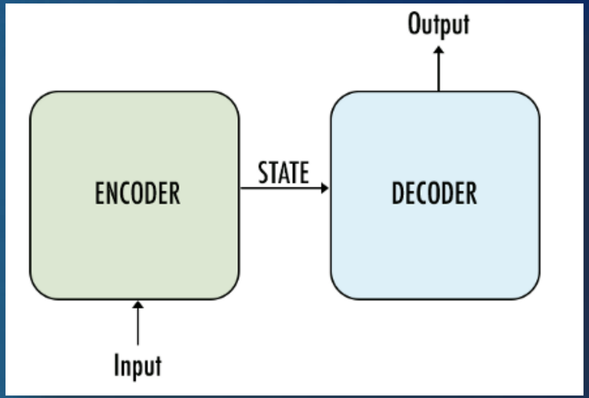
- Transformer-based models (e.g. OpenAI's GPT (Generative Pre-trained Transformer))
  - Designed to understand the context in sequences, like sentences or paragraphs. They use a mechanism called "self-attention" to determine which words in a sentence are important to understand the meaning, allowing them to handle tasks like translation or text generation effectively.
- Generative Adversarial Networks (GAN) - (e.g. DeepArt, StyleGAN by NVIDIA)
  - Made up of two AI models that work against each other: one creates new data (like fake images), and the other tries to detect if the data is real or fake. This competition helps the model get better at generating realistic data over time.
- Variational Autoencoders (VAEs) - (e.g. DALL-E by OpenAI, Jukebox by OpenAI)
  - Models that learn to compress data (like images) into a simpler form and then reconstruct the original data from this compressed version. They add a bit of randomness to the compression process, allowing them to create new, similar data by sampling from this simplified representation.
- Multimodel Models (e.g. CLIP by OpenAI)
  - Designed to understand and generate content that involves multiple types of data, like text and images together. They learn from both types of data at the same time, enabling them to perform tasks like describing an image in words or finding an image that matches a description.



Source: https://www.jaroeducation.com/blog/future-of-generative-ai/

Transformer-based Models

- Transformer-based models are a type of deep learning model architecture that uses a mechanism called "self-attention" to process data, particularly sequences like text.
- At a high level, a transformer model consists of an **encoder** and a **decoder**. The encoder converts input text into an intermediate representation and passes it to the decoder, the decoder converts that intermediate representation into useful text.
- A Large Language Model (LLM) is a large-scale transformer-based neural network that excels in understanding and generating human language
- LLMs are a specific type of transformer-based model that are particularly designed for understanding and generating human language.
- The "large" in LLM refers to the size of the model, which is typically very big (often with billions of parameters), allowing it to learn from massive amounts of text data.
- ChatGPT, Google Bard, Microsoft Copilot and Claude are examples of LLMs

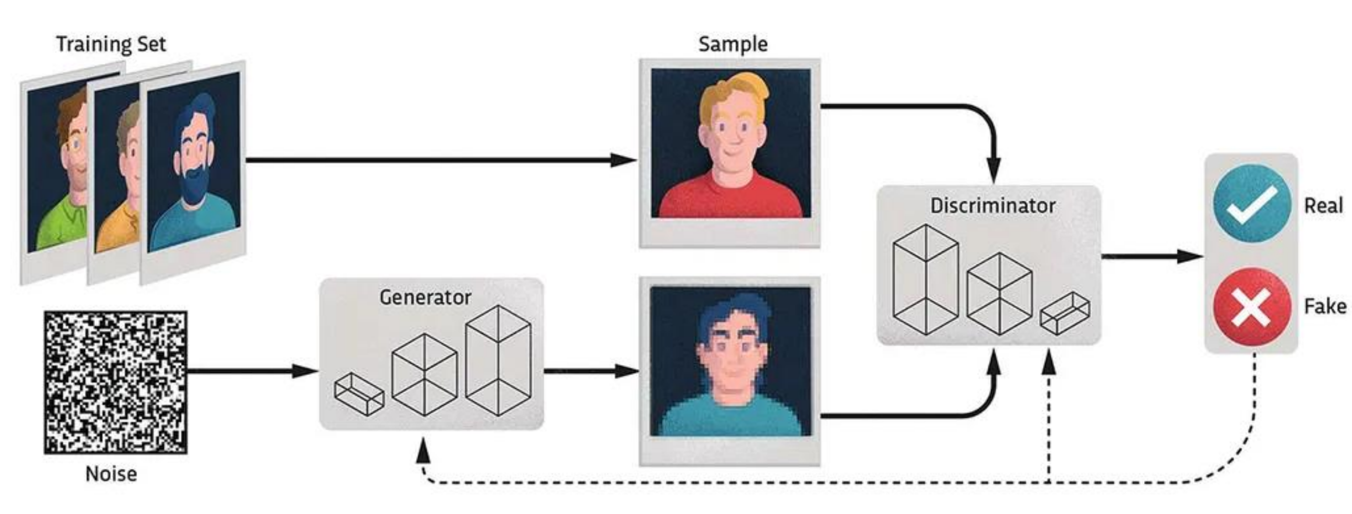


Source: https://www.tredence.com/generative-ai-101

Generative Adversarial Network

- GANs (generative adversarial networks) are clever machine learning (ML) algorithms that use neural networks (simplified computer models of the brain) in a specific way
- They are called 'generative' because once they have been trained on a dataset, they can generate new examples that resemble what they've seen. Train a generative ML algorithm on a dataset comprising millions of photos of faces and it will be able to generate new photorealistic faces

The **generator** creates new data (like images or text) by trying to mimic the real data it has been trained on. Think of it like an artist trying to paint a picture that looks like a real photograph



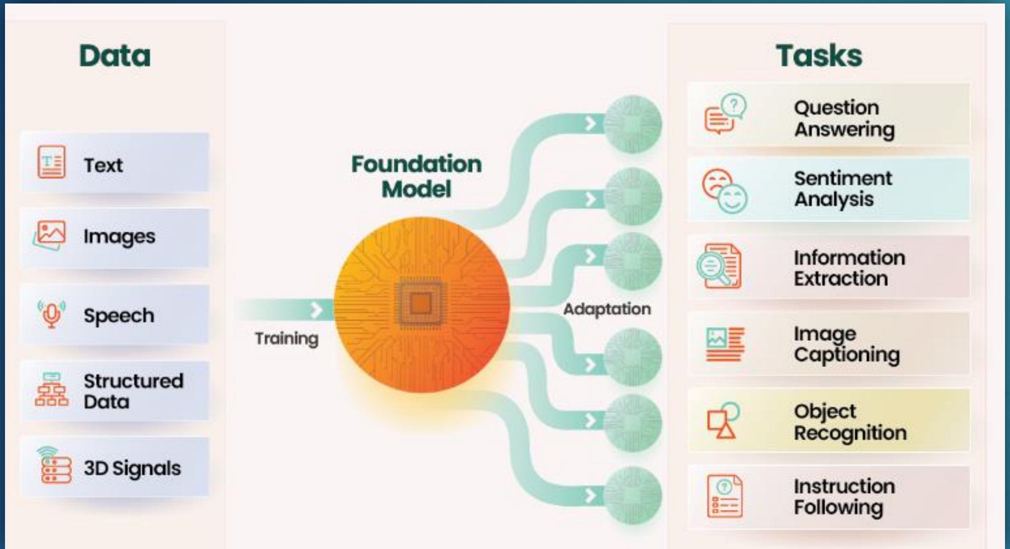
The **discriminator**, tries to tell whether the data it receives is real (from the training set) or fake (created by the generator). It's like a judge who decides whether the picture is a real photograph or a fake one.

- The **generator** and **discriminator** play a game where the generator tries to create better and better fakes, while the discriminator tries to get better at spotting them.
- Over time, the generator improves its ability to create realistic data, and the discriminator improves its ability to distinguish between real and fake data

Source: https://www.sciencefocus.com/future-technology/how-do-machine-learning-gans-work

Foundation Models

- A foundation model in GenAI is a large, pre-trained AI model that has learned to understand a wide range of patterns and information from massive amounts of data
- They are often built using transformers due to their flexibility and power in understanding context. GANs, VAEs, and multimodal models are more specialized, but their techniques can be integrated into foundation models to enhance capabilities in generating specific types of data or handling multiple data types.
- Foundation models include OpenAI's GPT-4, Google's Gemini, Meta's Llama 2, and Anthropic's Claude

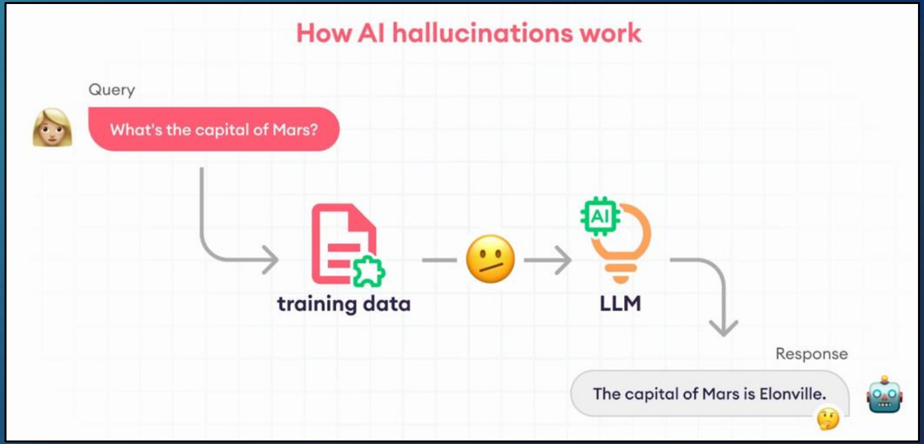


- **GPT-4** is the underlying large language model, capable of a wide range of applications beyond just conversation
- **ChatGPT** is a conversational AI application that uses models like GPT-4, tailored for dialogue and designed to interact with users in a chat format
- While all LLMs are foundation models focused on text, not all foundation models are LLMs
- Foundation models have a broader scope and can handle multiple data types, whereas LLMs are specifically designed for language-related tasks

Source: https://www.linkedin.com/pulse/comparing-generative-ai-services-aws-azure-gcp-dr-rabi-prasad-3zqdc/

GenAI Hallucinations

- Occur when models generate responses that sound plausible but are factually incorrect or not reality-based
- Arise from gaps or conflicts in training data, leading to content that appears accurate but is misleading
- Why is this a problem?
  - Can damage reputations if false or defamatory content is produced.
  - May cause medical errors if relied upon in healthcare decisions.
  - Risk legal and financial penalties if private data is exposed.
  - Undermines trust by spreading convincing but false information, fueling misinformation and "fake news."
- Cloud providers now offer guardrails and content filters to reduce hallucination risks



Source: https://hackernoon.com/gen-ai-hallucinations-the-good-the-bad-and-the-costly

GenAI and Cloud Migration

- Remember this discussion topic?
- The ACME Company, which is a major information provider for News, Financial and Legal content has decided to make a major move to the cloud. They are currently not using any cloud providers.
- As a cloud consulting group, you have been hired to provide ACME with ideas on how they can achieve this goal
- What if we gave this problem to AI?

Source: https://www.computerweekly.com/news/366555296/How-generative-AI-and-cloud-complement-each-other

GenAI and Cloud Migration

Discovery & Planning

Documenting Current Architectures

- Scan environments to construct topological maps of servers, apps, databases
- Generate interactive visualizations and documentation of existing infrastructure

Optimizing Target Architectures

- Analyze dependencies and usage patterns to identify consolidation and modernization opportunities
- Recommend optimized partitioning of systems and data for cloud

Cloud Readiness

Refactoring Codebases

- Migrate legacy code to modern languages better suited for cloud platforms
- Decompose monoliths into distributed microservices applying cloud design patterns

Migrating Data

- Profile legacy databases and extract schemas
- Generate ETL logic to transform and load data into cloud data platforms

Migration & Validation

Automating Deployment

- Generate infrastructure-as-code definitions for provisioning target cloud environments
- Package and deploy applications with their transformed code and data to the cloud

Validating Correctness

- Auto-generate unit, integration and performance tests for migrated apps
- Validate functionality, scalability and compliance with pre-migration configurations

Optimization & Modernization

Incremental Modernization

- Prioritize and schedule incremental modernization of remaining legacy processes
- Leverage AI to convert legacy systems and data incrementally post-migration

Optimizing Cost and Utilization

- Continuously optimize right-sizing of cloud resources based on actual usage data
- Notify on migration elements ripe for modernization to prevent cloud waste

GenAI

Source: https://medium.com/@bijit211987/accelerating-cloud-migration-efficiency-with-generative-ai-fdc7eaea1509

How does the Cloud Support GenAI?

- Scalable Compute Resources
  - GenAI models require substantial computational power for training and inference. Public cloud providers offer scalable compute resources, allowing users to access powerful GPUs and TPUs on-demand
- Large-Scale Data Storage
  - Training GenAI models involves vast amounts of data. Public clouds provide extensive storage solutions that can handle the enormous datasets required
- Cost Efficiency
  - With a pay-as-you-go model, organizations can scale their AI operations flexibly, avoiding the high upfront costs associated with building and maintaining on-premises infrastructure
- Rapid Experimentation and Deployment
  - Cloud environments support rapid provisioning of resources, allowing for quick experimentation with different model architectures
- Managed Services and Tools
  - Cloud providers offer managed services and tools for machine learning and AI, such as automated machine learning (AutoML), pre-built AI models, and integrated development environments

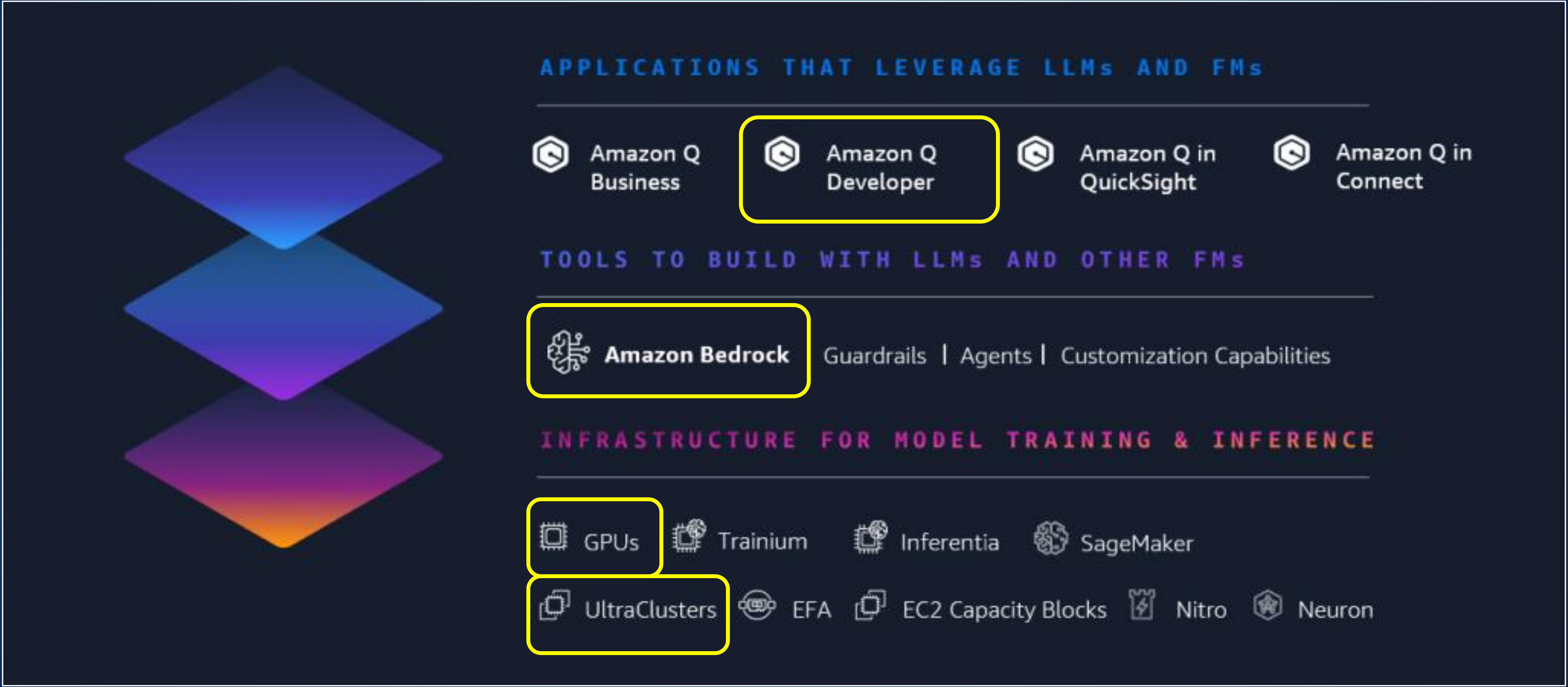
Source: https://blogs.idc.com/2024/07/31/embracing-the-future-how-genai-is-revolutionizing-cloud-infrastructure-for-growing-tech-vendors/

Public Cloud offerings of GenAI Services

GenAI	Comparison of Google, Azure & AWS GenAI Services			
Hyperscaler	**Build & Deploy**	**Foundational Models**	**PrepareData**	**AI Infrastructure**
Google	VertexAI	PaLM (Language)	Data Set	supports NVIDIA L4 and H100 Tensor Core GPUs, as well as prior GPU generations including A100 and more
	Generative AI Studio - GUI for Models	Codey (Code)	Data (Human) Labelling	
	ColabEnterprise & Workbench Notebooks	ImageGen(Image)	FeatureStore	
	ModelGarden - Model Repository	Chirp (Translation) Dialogflow (conversations)	Google Marketplace offerings	
AWS	AWS Bedrock	AWS Titan	Amazon Redshift, Amazon RDS, Snowflake, Amazon Athena, and AWS Glue	AWS Trainium, AWS Inferentia, and NVIDIA GPUs.
	Huggingface	Vision & Recommendation	Third party Foundation models - Cohere's & Claude 2 etc.	
	SageMaker (Build Models)	Amazon Lex for building chatbots.		
	Amazon Comprehend - Find insights and relationships in text	Rekognition - Video & Image Third party Models	Vector engine for Amazon OpenSearch Serverless	
		Amazon Polly for text-to-speech,		
Azure	Azure OpenAI	GPT-4 models	Azure Machine Learning - Build, deploy, and manage models	ND H100 v5 Virtual Machine series, equipped with NVIDIA H100 Tensor Core graphics processing units (GPUs) and NVIDIA H100 GPUs interconnected by NVIDIA Quantum-2 InfiniBand networking.
	Azure AI Studio	Azure Bot Service for building chatbots		
	Copilots	DALL-E models	Microsoft Fabric - Unify, Manage & Govern Data	
By Ravi Saraswathi	Prompt Engineering	Azure Cognitive Services for computer vision	Azure App Services	

Source: https://www.linkedin.com/pulse/comparing-generative-ai-services-aws-azure-gcp-dr-rabi-prasad-3zqdc/

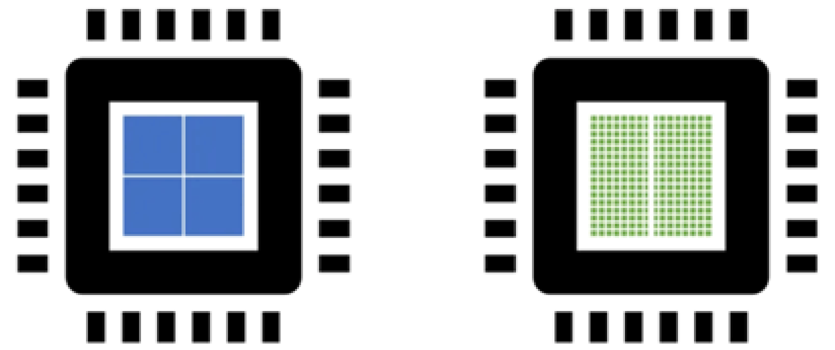
AWS GenAI Tools and Services



Source https://aws.amazon.com/blogs/awsmarketplace/generative-ai-partner-offerings-in-aws-marketplace-core-infrastructure-software/

Infrastructure

- GenAI training requires massive parallel processing of data from many sources, with thousands of computations happening simultaneously
- Regular CPU-based servers can't handle this training load, so that's where GPUs come in
- A GPU (Graphics Processing Unit) is a specialized processor originally designed to accelerate the rendering of graphics and images
- They have evolved to handle complex mathematical computations, making them highly efficient for parallel processing tasks, like GenAI



CPU	GPU
Central Processing Unit	Graphics Processing Unit
4-8 Cores	100s or 1000s of Cores
Low Latency	High Throughput
Good for Serial Processing	Good for Parallel Processing
Quickly Process Tasks That Require Interactivity	Breaks Jobs Into Separate Tasks To Process Simultaneously
Traditional Programming Are Written For CPU Sequential Execution	Requires Additional Software To Convert CPU Functions to GPU Functions for Parallel Execution

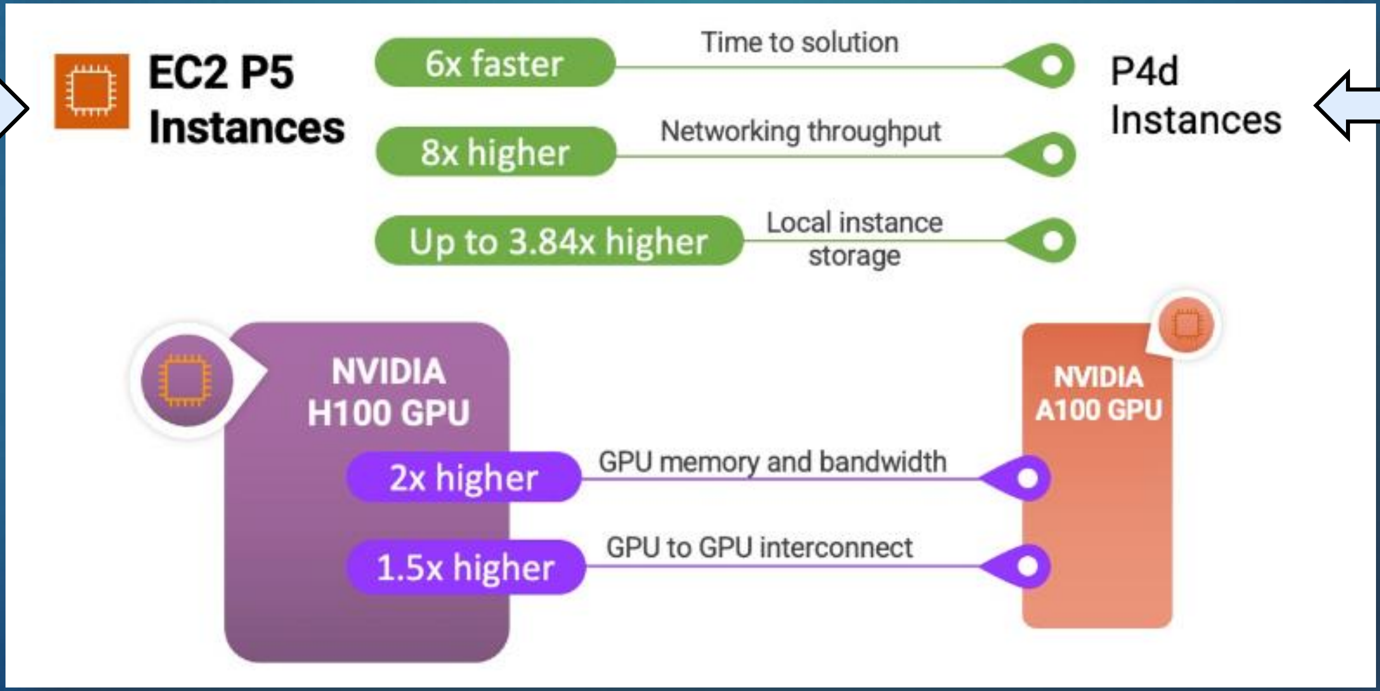
Source: https://www.cherryservers.com/blog/what-is-gpu-computing

Infrastructure - AWS

- EC2 P5 Instances powered by the latest NVIDIA H100 Tensor Core GPUs and will provide a reduction of up to 6 times in training time

Instance Size	vCPUs	Memory (GiB)	GPUs (H100)	Network Bandwidth (Gbps)	EBS Bandwidth (Gbps)	Local Storage (TB)
p5.48xlarge	192	2048	8	3200	80	8 x 3.84

2023

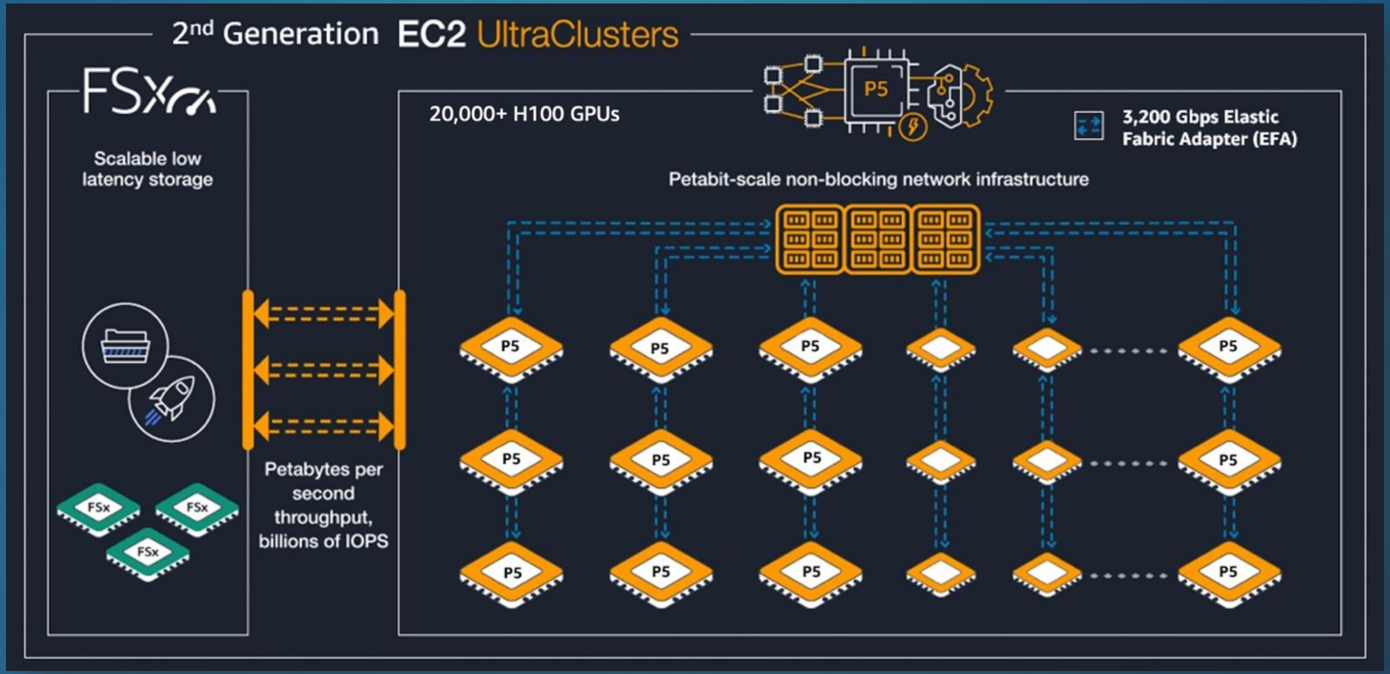


2022

Source: https://aws.amazon.com/blogs/aws/new-amazon-ec2-p5-instances-powered-by-nvidia-h100-tensor-core-gpus-for-accelerating-generative-ai-and-hpc-applications/

Infrastructure – AWS EC2 UltraClusters

- EC2 UltraClusters consist of thousands of accelerated EC2 instances that are co-located in an AWS Availability Zone and interconnected using Elastic Fabric Adapter (EFA) networking in a petabit-scale nonblocking network



- They have local storage in the P5 nodes and access to the FSx implementation of the Lustre parallel file system, and which can deliver petabytes per second throughput, billions of IOPs

Source: https://aws.amazon.com/blogs/aws/new-amazon-ec2-p5-instances-powered-by-nvidia-h100-tensor-core-gpus-for-accelerating-generative-ai-and-hpc-applications/

AWS Bedrock

- Fully managed service for deploying and integrating generative AI models
- Provides API access to multiple foundation models without managing infrastructure
- Supports customization and fine-tuning for business-specific use cases
- Scales easily from small prototypes to enterprise workloads
- Key features:
  - Managed infrastructure to reduce operational overhead
  - Integration with AWS services like SageMaker, Lambda, and S3
  - Access to models from Anthropic, Cohere, Meta, Stability AI, AI21 Labs, Mistral, and Amazon Titan
  - Enterprise-grade security with encryption and IAM controls

Amazon Bedrock

How does Amazon Bedrock work?

- Provides easy access to foundation models (FMs) from providers including AI21 Labs, Anthropic, Cohere, Meta, Mistral AI, Stability AI, and Amazon
- Uses a single API to integrate multiple models, making it simple to switch or upgrade to the latest versions with minimal code changes
- Enables organizations to quickly adopt new GenAI innovations without managing infrastructure or rewriting applications

Source: https://www.forbes.com/sites/moorinsights/2023/08/04/aws-turbocharges-foundation-models-with-smart-ai-agents/

Amazon Titan

- Amazon Titan is a family of foundation models available through AWS Bedrock, designed for secure and scalable AI deployment
- Titan models integrate seamlessly with AWS services to support a wide range of generative AI use cases
- Mode Types:
  - Text Premier – High-performance text generation, summarization, and Q&A (e.g., chatbots)
  - Image Generator – Creates high-quality images from text prompts for industries like advertising, e-commerce, and media
  - Text Embeddings – Converts text into numerical vectors for semantic search, clustering, and contextual relevance (e.g., product recommendations)

Source: https://blogs.idc.com/2024/07/31/embracing-the-future-how-genai-is-revolutionizing-cloud-infrastructure-for-growing-tech-vendors/

Amazon Q Developer

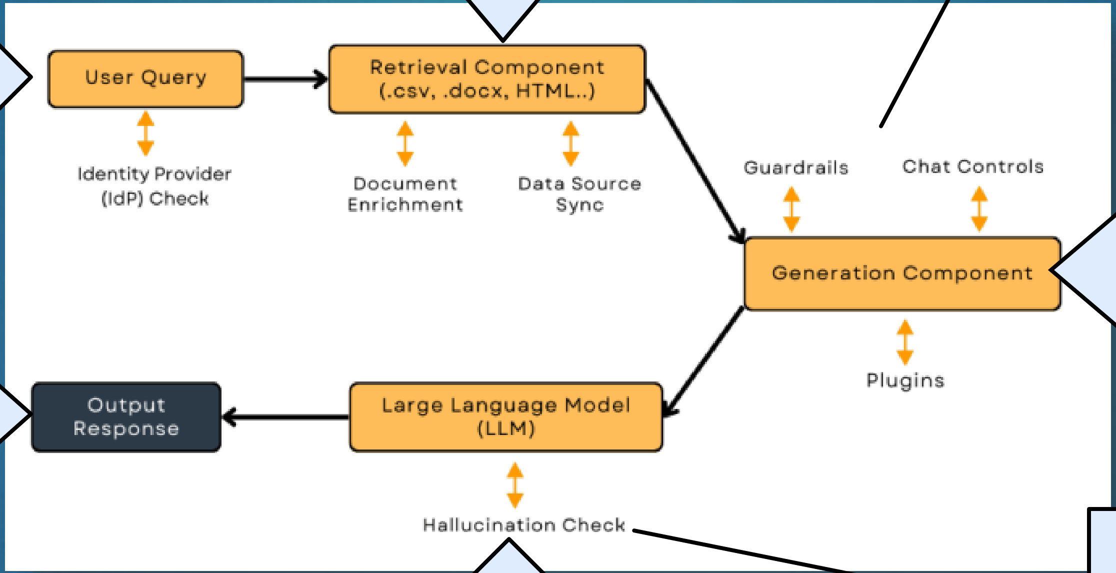
- Amazon Q Developer is a GenAI-powered assistant that helps you understand, build, extend, and operate AWS applications
- Integrated with IDEs like Visual Studio and IntelliJ, it provides real-time software development assistance.
- Capabilities include:
  - Conversational code assistance
  - Inline code completions and new code generation
  - Security scans, debugging, and language/framework upgrades
  - Available through a Free Tier and powered by Amazon Bedrock.

Amazon Q Developer – How does it work?

Upon receiving the user query, Q's retrieval component swings into action. It searches through various indexed documents to find those most relevant to the query. These documents could be in various formats, such as .csv, .docx, etc.

These tools allow administrators to define what topics Q can discuss and the sources it can use for generating responses, ensuring the conversation stays within predefined boundaries.

User submits a query that could range from simple informational requests to complex problem-solving tasks, depending on the user's needs.



This component, leveraging a large language model (LLM), synthesizes the information from these documents to construct a coherent and relevant response. The LLM is trained on a vast corpus of data, enabling it to generate accurate and contextually appropriate answers.

The final output is a response that is presented back to the user. This response aims to be clear, accurate, and directly relevant to the initial query, providing the user with the information or guidance they sought.

Before delivering the response, Q performs a **hallucination check**. This step is crucial to ensure that the response is based on actual data and knowledge, avoiding any irrelevant or speculative content that doesn't align with the provided documents or user query context.

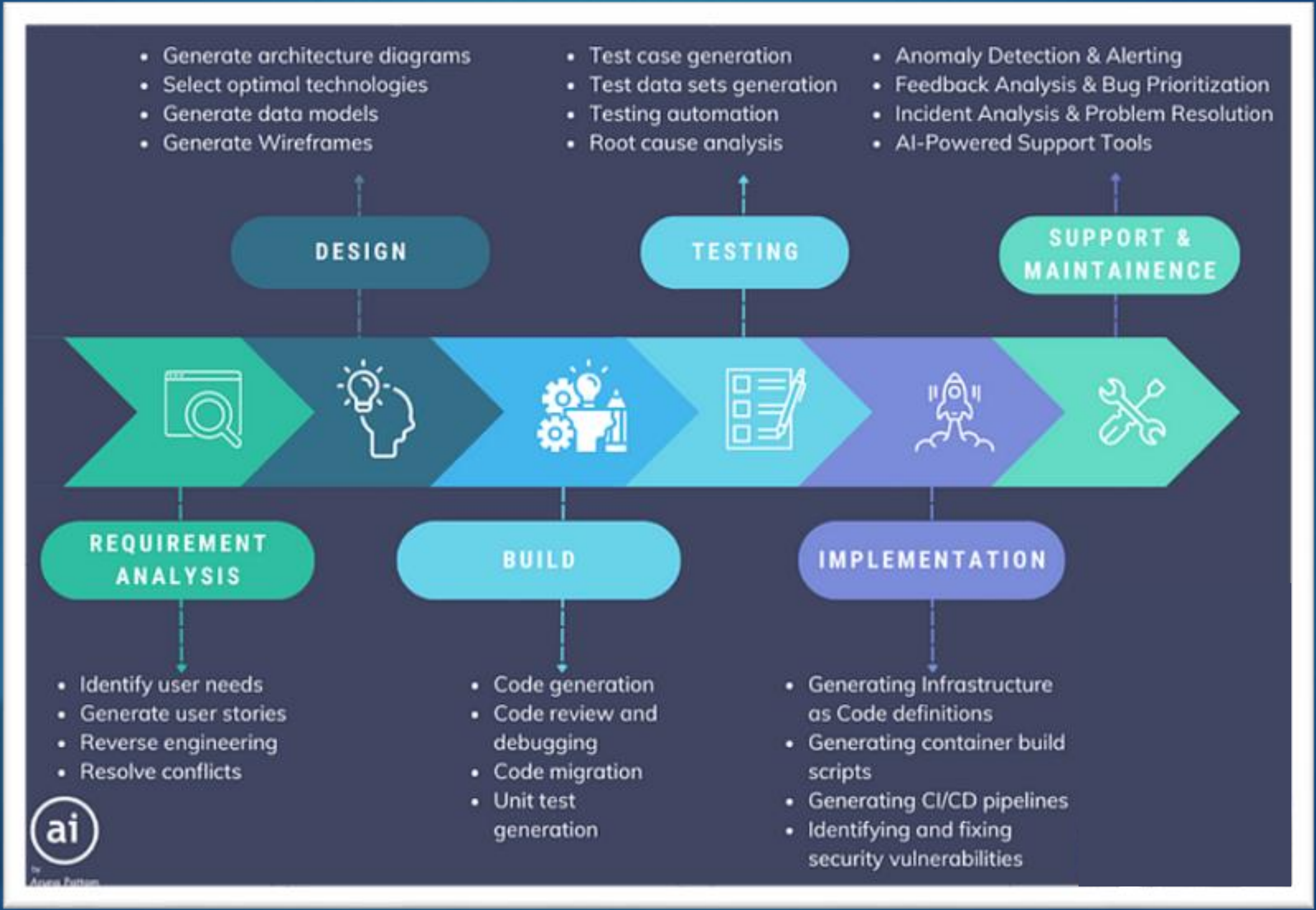
This step involves the LLM processing the information to formulate an answer. The model's training enables it to understand complex queries and provide detailed responses, drawing from the information contained in the retrieved documents.

Source: https://blog.economize.cloud/amazon-q-pricing/

AI Concerns and Ethics

Misinformation and Disinformation	• GenAI models can "hallucinate" produce content that appears credible but is factually incorrect, misleading, or entirely fabricated.
Bias and Fairness	• Biased AI outputs can lead to unfair treatment, discrimination, or reinforce harmful stereotypes, especially when used in applications like hiring, law enforcement, or content moderation
Privacy and Data Security	• Training GenAI models often involves using large datasets, which can include sensitive or personal information. There is a risk that these models could inadvertently memorize and output this private information.
Intellectual Property and Copyright Infringement	• GenAI can create content that mimics or replicates existing works, potentially infringing on intellectual property rights. There are also concerns about the use of copyrighted data during the model training process without proper consent or compensation.
Accountability and Transparency	• The decision-making processes of GenAI models are often opaque, making it difficult to understand how outputs are generated. This "black box" nature of AI can create challenges in assigning accountability for errors or harmful outcomes
Autonomy and Human Control	• As GenAI systems become more advanced, there is a concern about the loss of human control over AI-driven decisions and actions, particularly in high-stakes areas such as autonomous vehicles, military applications, or healthcare
Environmental Impact	• Training large GenAI models requires significant computational power, which can have a substantial environmental footprint due to high energy consumption

GenAI in Software Development



Source: https://arunapattam.medium.com/generative-ai-in-software-engineering-fc7da0710050

GenAI in Software Development – Future?

- GenAI can lead to faster development with a 28% increase in task completion speed, which is expected to reach 54% by 2025
- Additionally, documenting code functionality for maintainability and writing new code can be accomplished in 50% less time
- Developers using GenAI-based tools for complex tasks had a 25% to 30% higher likelihood of completing the tasks
- Finally, using AI tools for testing processes resulted in a 31% increase in completion, and this is estimated to rise to 57% in 2025

Source: https://lab45thinktank.com/article/reimagining-software-engineering-with-genai-a-primer/

- Within your team, discuss the following:
  - What concerns or excites you about GenAI?
  - Any interesting tools/technologies you'd like to share?
  - Where do you see GenAI most prominent in the next 5-10 years?
- Teams have 10 minutes to share their responses